Accessible Quantum Natural Gradient: Readout-Induced Geometry for Variational Optimization

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Quantum natural gradient (QNG) preconditions variational optimization with the full quantum Fisher geometry, although a learning model typically interacts with the state through a fixed measurement and a restricted retained readout. We introduce the accessible quantum natural gradient (AQNG), which instead uses the pullback of the geometry exposed by that measurement–readout interface. For commuting retained outcome features, the accessible metric $G_{\mathrm{acc}} = B^{-1} \sum_b J_b^{\mathsf{T}} \Sigma_b^{+} J_b$ is the exact Gram matrix of the Fisher-score projection onto the retained feature span. To isolate readout orientation from rank, we compare physical low-weight readouts with equal-rank Haar-random and cross-fitted tangent-aligned readouts. A frozen campaign of 470 experiment cells (3160 terminal method rows) covers four binary datasets, four circuit families, qubit and depth scaling, readout-order ablations, and finite-shot training; a separate 15-cell follow-up adds matched SGD and Adam baselines in the larger/deeper SU(2)-Haar settings. Across the 240 generic primary cases, aligned AQNG has the lowest mean test loss (0.5945) and highest mean accuracy (0.7947) among eight methods, while random AQNG and SGD remain close. In the pooled larger/deeper follow-up, aligned AQNG beats SGD in 15/15 matched cases (Holm-adjusted $p = 1.22 \times 10^{-3}$) and full QNG in 14/15 ($p = 5.80 \times 10^{-3}$), but its advantage over Adam is not statistically resolved. Geometry diagnostics show that random readouts track the rank baseline, whereas normalized aligned retention can grow sharply with system size. More accessibility is nevertheless not synonymous with better trainability: higher-order readouts can increase retained tangent mass while worsening conditioning and task loss, and a number-conserving $U(1)$ control retains nearly all accessible tangent mass while becoming chance-like as the descent signal collapses. AQNG therefore provides an interpretable, measurement-aware preconditioner, but accessible tangent geometry is not a universal proxy for task usefulness.

I. INTRODUCTION

The quantum Fisher information matrix (QFIM) equips a parameterized quantum state with a local information geometry and underlies the quantum natural-gradient update [1, 2]. This geometry is measurement independent: it quantifies infinitesimal distinguishability available in the state before a particular experimental interface is fixed. Variational quantum learning, by contrast, is driven by a classical record obtained from a chosen measurement and usually compressed further to a restricted family of observables, correlators, marginals, or task features. The optimizer therefore need not have operational access to every direction represented by the full QFIM.

Our earlier work separated these layers geometrically. For a fixed measurement, normalized tangent-score vectors define a centered score space with covariance C, while a retained rank-r readout defines a projector P. The retained tangent mass

$$R = \mathrm{Tr}(PC), \qquad \rho = \frac{R}{r/N} \tag{1}$$

compares the actual readout orientation with the rank-only random-subspace baseline r/N [3]. Rank fixes the null accessibility scale, the spectrum of C fixes the width of random-orientation fluctuations, and the relative orientation of the readout and tangent eigenspaces fixes the observed retention. Equal-rank readouts can therefore expose very different fractions of the same measured tangent geometry.

Restricted measurement geometry has also appeared from a representational viewpoint. Gross and Rieser analyze quantum extreme learning machines through Pauli-transfer matrices and characterize which encoded Pauli features remain decodable from a chosen measurement row space [4]. Their setting is a fixed quantum reservoir followed by a trained classical readout; the relevant object is the decodability of encoded input features after channel evolution. The present setting is differential rather than representational. We study parameter tangents of a trainable variational circuit after a measurement has been fixed, Fisher-normalize the resulting score geometry, and ask how that accessible geometry should precondition parameter updates. Thus the common linear-algebraic skeleton—projection onto a retained measurement subspace—is used here for a distinct optimization problem.

This distinction motivates the accessible quantum natural gradient (AQNG). Let $f(x)$ denote a vector of retained commuting outcome features, $J = \partial_\theta \langle f \rangle$ their parameter Jacobian, and Σ their outcome covariance. The matrix

$$G_{\mathrm{acc}} = \frac{1}{B} \sum_{b=1}^{B} J_b^{\mathsf{T}} \Sigma_b^{+} J_b \tag{2}$$

is the pullback of the Fisher-score projection onto the retained feature span. AQNG uses this interface-induced metric to precondition the supervised loss gradient. It is not introduced as a low-rank numerical approximation whose objective is to reproduce the full QFIM. Instead, it asks whether the geometry actually exposed by a fixed readout is sufficient, useful, and better conditioned for the task at hand.

Three issues must then be separated experimentally. First, *rank* and *orientation*: a physical readout should be compared with equal-rank random and tangent-aligned controls. Second, *accessibility* and *trainability*: retaining a large fraction of tangent information need not imply that the supervised gradient is informative. Third, *geometric quality* and *optimizer performance*: a readout that maximizes tangent retention can still yield an ill-conditioned metric or a poorer task trajectory. These distinctions are especially important in discussions of barren plateaus, where rescaling an already unresolvable gradient is not equivalent to restoring trainability [5].

We evaluate these questions in a frozen experiment matrix containing 470 paper-scale cells and 3160 terminal method rows, spanning four binary datasets, four circuit families, qubit and depth scaling, readout-order ablations, and finite-shot training [6]. The primary benchmark compares eight optimizers under matched initialization and data partitions. Across the 240 nonconserving dataset–architecture cases, aligned AQNG has the lowest pooled mean test loss and highest pooled mean accuracy, although random AQNG and SGD remain close and no universal winner emerges. A targeted larger/deeper SU(2)-Haar follow-up adds matched Adam and SGD baselines: pooled over 15 cells, aligned AQNG significantly outperforms SGD and full QNG after Holm correction, while the difference from Adam is unresolved. The geometry diagnostics are equally important. Random rank-matched readouts remain near the rank baseline, cross-fitted alignment increases retained tangent mass, higher-order readouts can increase retention while degrading conditioning and task loss, and the $U(1)$ control remains highly accessible while its effective descent signal collapses with system size. The resulting claim is deliberately narrower than "AQNG is always a better optimizer": measurement-accessible geometry is a controllable and interpretable optimization resource, but it is not identical to task usefulness.

II. ACCESSIBLE TANGENT GEOMETRY AND AQNG

A. Fixed measurement, retained features, and score geometry

Consider a fixed measurement with outcomes $z \in \{1,\ldots,K\}$ and probability vector $p_{\boldsymbol{\theta}}(x)$ for input x. Let

$$Q = \frac{\partial p_{\boldsymbol{\theta}}(x)}{\partial \boldsymbol{\theta}} \in \mathbb{R}^{K\times d}, \qquad D_p = \mathrm{diag}(p_{\boldsymbol{\theta}}(x)). \tag{3}$$

On the regular support of p, the Fisher-normalized probability tangent matrix is

$$S = D_p^{-1/2} Q, \tag{4}$$

and the classical Fisher information matrix (CFIM) of the complete measurement record is

$$G_{\mathcal{M}} = S^{\mathsf{T}} S = Q^{\mathsf{T}} D_p^{+} Q. \tag{5}$$

For a pure-state model, $G_{\mathcal{M}} \preceq G_Q$, with equality only when the chosen measurement preserves the relevant quantum Fisher geometry.

The learning interface generally retains only a collection of commuting outcome functions $f_1(z),\ldots,f_r(z)$. Write $F \in \mathbb{R}^{K\times r}$ for the matrix with entries $F_{z\alpha} = f_\alpha(z)$, let

$$\boldsymbol{\mu} = F^{\mathsf{T}} p, \qquad \widetilde{F} = F - \mathbf{1}\boldsymbol{\mu}^{\mathsf{T}}, \tag{6}$$

and define the feature covariance and parameter Jacobian

$$\Sigma = \widetilde{F}^{\mathsf{T}} D_p \widetilde{F}, \qquad J = \frac{\partial \boldsymbol{\mu}}{\partial \boldsymbol{\theta}} = F^{\mathsf{T}} Q. \tag{7}$$

Because probability tangents satisfy $\mathbf{1}^{\mathsf{T}} Q = 0$, one also has $J = \widetilde{F}^{\mathsf{T}} Q$.

B. Exact score-projection identity

Define $B = D_p^{1/2}\widetilde{F}$. Then $\Sigma = B^{\mathsf{T}} B$ and $J = B^{\mathsf{T}} S$. Therefore

$$G_{\mathrm{acc}} \equiv J^{\mathsf{T}} \Sigma^{+} J \tag{8}$$

$$= S^{\mathsf{T}} B (B^{\mathsf{T}} B)^{+} B^{\mathsf{T}} S \tag{9}$$

$$= S^{\mathsf{T}} P_B S, \tag{10}$$

where P_B is the Euclidean projector onto the covariance-weighted retained feature span. Thus, for commuting features derived from a common measurement record, $J^{\mathsf{T}}\Sigma^{+}J$ is not merely a heuristic approximation to the CFIM: it is the exact Gram matrix of the measurement score after projection onto the retained readout subspace. Consequently,

$$0 \preceq G_{\mathrm{acc}} \preceq G_{\mathcal{M}} \preceq G_Q. \tag{11}$$

Rank deficiency is handled by the Moore–Penrose pseudoinverse. In finite-shot experiments we additionally regularize the empirical covariance; that regularized metric should be read as a stabilized estimator of Eq. (10), not as a new exact projector identity.

This construction also clarifies the scope of the method. A list of noncommuting observables does not by itself define a single jointly measurable covariance matrix Σ; one must first specify a measurement grouping or another explicit experimental interface. The present experiments use diagonal commuting outcome functions derived from computational-basis samples.

C. Accessible quantum natural gradient

For a metric minibatch B_G, AQNG averages the accessible pullbacks,

$$G_A(\boldsymbol{\theta}) = \frac{1}{|B_G|} \sum_{x \in B_G} J_x^{\mathsf{T}} \Sigma_x^{+} J_x. \tag{12}$$

The supervised loss gradient $g = \nabla_{\boldsymbol{\theta}} L$ is shared across optimizer comparisons. AQNG forms a controlled preconditioned direction

$$d_A = (\widehat{G}_A + \lambda_{\text{eff}} I)^{-1} g, \qquad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta d_A, \tag{13}$$

where $\widehat{G}_A$ denotes the chosen metric normalization and λ_{eff} the damping convention. The implementation supports trace or maximum-eigenvalue normalization, absolute or relative damping, and both Euclidean and accessible-metric trust controls. The frozen paper-scale protocol uses trace normalization with absolute damping.

The full-QNG baseline instead uses the pure-state QFIM with the explicit convention

$$(G_Q)_{ij} = 4\,\mathrm{Re}[\langle \partial_i \psi | \partial_j \psi \rangle - \langle \partial_i \psi | \psi \rangle \langle \psi | \partial_j \psi \rangle]\,. \tag{14}$$

AQNG is therefore not defined by the requirement $G_A \approx G_Q$ in matrix norm. It intentionally changes the geometry by discarding score directions absent from the retained measurement interface.

D. Rank-matched readout orientation

The retained feature span is calibrated once at the common initial parameter point and then frozen during supervised training. The paper-scale controls use three equal-rank designs: a physical low-weight diagonal readout, a Haar-random rank-matched subspace of the same score space, and a cross-fitted aligned subspace estimated from independent label-free tangent directions. The aligned projector is evaluated on held-out tangents so that its reported retention is not the same-sample Ky Fan optimum.

For a normalized held-out tangent covariance C in a centered score space of dimension N and a rank-r projector P, we report

$$R = \mathrm{Tr}(PC), \qquad \rho = \frac{R}{r/N}. \tag{15}$$

A Haar-random rank-r subspace has $\mathbb{E}\rho = 1$; deviations of the physical or aligned readout from this baseline diagnose orientation rather than rank. Crucially, R and ρ are geometry diagnostics, not supervised performance metrics. The experiments below test when better tangent accessibility does—and does not—translate into a better optimization trajectory.

III. EXPERIMENTAL DESIGN

A. Benchmarks and circuits

We use four binary classification benchmarks: Iris, Breast Cancer, Wine, and Digits. Each experiment uses a fixed train/test split per dataset while the stochastic circuit/training seed is varied. The benchmark configuration uses six qubits, 80 samples, three variational layers, and 20 training steps. Four circuit families are compared: `ryrz_cz`, `su2_cnot`, `su2_haar`, and the number-conserving `u1_rzxy`. The $U(1)$ family is retained as a symmetry-constrained control because previous accessibility calculations show that its tangent orientation differs strongly from generic isotropic behavior [3**?**].

B. Optimization protocol

AQNG and full QNG use learning rate $\eta = 0.03$ and damping $\lambda = 10^{-3}$. The loss minibatch contains eight examples, the metric minibatch contains four, and the metric is refreshed every two optimization steps. Each dataset–architecture cell contains 20 paired seeds, yielding $4 \times 4 \times 20 = 320$ paired comparisons and 640 terminal method rows. The initial loss gradient is shared within each pair; the recorded initial-gradient cosine is numerically unity.

C. Statistics and timing

For each cell we report mean test loss and accuracy over 20 seeds. Paired loss differences are $\Delta L = L_{\text{AQNG}} - L_{\text{QNG}}$, so negative values favor AQNG. Two-sided Wilcoxon signed-rank tests are corrected across the 16 cells by Holm's procedure. Wall-clock measurements refer only to the implemented simulator stack. The AQNG and full-QNG metric paths use different simulator backends in the present implementation while the loss path is shared; timing is therefore an implementation result, not an intrinsic quantum-resource claim.

IV. RESULTS

A. Predictive performance

Excluding the symmetry-constrained $U(1)$ control, AQNG improves mean test accuracy in nine of twelve dataset–architecture cells (Table I and Fig. **??**). The result is not universal: RyRz-CZ favors full QNG on Iris and Digits, and SU(2)-CNOT is essentially tied on Breast Cancer. This heterogeneity supports an architecture-dependent accessible-geometry picture rather than a claim of universal optimizer dominance.

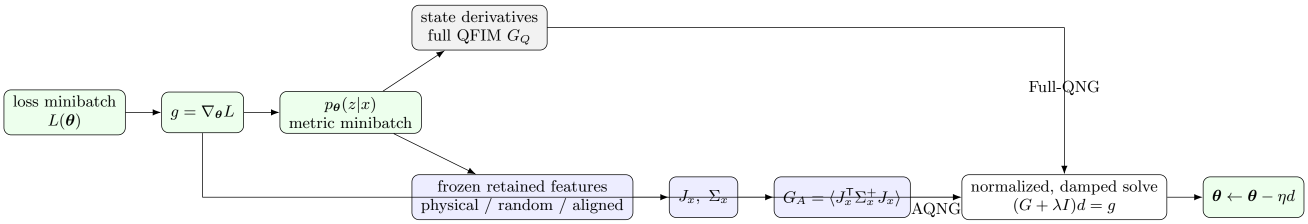

FIG. 1. Geometric and algorithmic layers. The loss gradient is shared across methods. AQNG first fixes a measurement/readout interface, projects the measurement score onto the retained feature span through Eq. (10), and pulls that geometry back to parameter space. Full QNG instead uses the full state-space QFIM. Physical, random, and cross-fitted aligned AQNG variants differ only in the frozen equal-rank readout subspace used to define the accessible metric.

TABLE I. Mean test accuracy over 20 paired seeds. Values are percentages. Δ is AQNG minus full QNG in percentage points.

Dataset	Architecture	AQNG	Full QNG	Δ
Iris	RyRz-CZ	70.75	75.75	-5.00
Iris	SU(2)-CNOT	86.50	79.25	+7.25
Iris	SU(2)-Haar	95.50	87.25	+8.25
Breast Cancer	RyRz-CZ	67.25	56.00	+11.25
Breast Cancer	SU(2)-CNOT	62.00	62.50	-0.50
Breast Cancer	SU(2)-Haar	68.50	58.75	+9.75
Wine	RyRz-CZ	69.75	52.25	+17.50
Wine	SU(2)-CNOT	71.25	63.50	+7.75
Wine	SU(2)-Haar	77.50	67.75	+9.75
Digits	RyRz-CZ	54.50	65.75	-11.25
Digits	SU(2)-CNOT	79.75	69.50	+10.25
Digits	SU(2)-Haar	81.75	77.50	+4.25

The strongest architecture-wide pattern is SU(2)-Haar: AQNG improves mean test accuracy on all four datasets. Its paired loss improvement remains significant after Holm correction on Iris ($p_{\mathrm{Holm}} = 9.84 \times 10^{-4}$) and Wine ($p_{\mathrm{Holm}} = 7.08 \times 10^{-3}$). Digits/SU(2)-CNOT is also significant after correction ($p_{\mathrm{Holm}} = 1.47 \times 10^{-3}$), with mean paired loss difference -0.232 and mean accuracy gain 10.25 percentage points. Breast Cancer/RyRz-CZ and Breast Cancer/SU(2)-Haar favor AQNG in mean loss and accuracy but do not remain significant after Holm correction. Figure **??** shows the complete terminal-loss pattern rather than only the significant cells.

B. Accessible Fisher mass and optimization utility

Successful AQNG behavior does not require retaining most of the full-QFIM trace. For SU(2)-Haar, the ≤ 2 accessible metric retains approximately 0.263, 0.262, 0.268, and 0.267 of QFIM trace on Breast Cancer, Digits, Iris, and Wine, respectively. The corresponding AQNG/full-QNG update-direction cosines are approximately 0.689, 0.668, 0.635, and 0.710. A full-QFIM trace measures total local state-space sensitivity, whereas the optimizer uses an inverse metric acting on a particular loss gradient. Removing Fisher mass poorly aligned with the retained learning interface need not remove useful preconditioning structure. Figure **??** visualizes the separation between retained Fisher mass and update alignment.

The nested diagnostics verify the Loewner hierarchy to floating-point tolerance for generic circuits. The largest residual in the complete run is 8.706×10^{-6} for Wine/$U(1)$, below the scale-aware 10^{-5} tolerance used by the pre-run smoke validation but above an inconsistent fixed 2×10^{-6} tolerance in the original aggregation script. Reaggregation changed no training data or statistics.

C. Symmetry-constrained control

The $U(1)$ architecture produces a qualitatively different regime. AQNG and full QNG obtain exactly the same mean test accuracy, with zero seedwise standard deviation, on all four tasks: 50% on Iris, 35% on Breast Cancer, 45% on Wine, and 50% on Digits. At the same time, mean AQNG/full-QNG direction cosines are 0.912, 0.858, 0.921, and 0.932. The failure therefore cannot be attributed simply to a poor approximation of the full natural-gradient direction.

This observation must be interpreted together with the preceding spectral theory. The earlier $U(1)$ result is not that low-order information is absent. Symmetry can instead concentrate tangent covariance and enhance alignment with low-weight readouts [3]. The present experiment establishes a different distinction: measurement accessibility is not identical to supervised-task discriminativeness. An accessible tangent direction can be large and can closely reproduce the full-QNG direction while still failing to separate labels under the chosen architecture, encoding, and readout. The $U(1)$ cells show small seed-consistent terminal-loss reductions under AQNG, but because accuracy is unchanged we do not interpret them as practical classification gains.

D. Implementation wall-clock

In the present simulator implementation, AQNG reduces mean wall-clock time by roughly 10.8–11.0× for RyRz-CZ, 38.2–38.7× for SU(2)-CNOT, 25.7–26.3× for SU(2)-Haar, and 35.4–36.1× for $U(1)$. These ratios are stable across datasets and are displayed in Fig. **??**. They show that the implemented accessible-metric path is substantially cheaper in this code base, but they do not establish a hardware-independent resource theorem.

V. DISCUSSION

A. From spectral accessibility to an optimizer

AQNG is the third step in a sequence. The isotropic readout-rank theory showed that state sensitivity can survive while a low-rank readout retains only a small fraction of visible score information [**?**]. The spectral theory generalized that result by replacing global isotropy with relative operator alignment $\mathrm{Tr}(TC_Q)$ [3]. AQNG makes this geometry operational by using the accessible tangent structure to precondition learning.

The empirical results support, but do not prove in full generality, the proposition that useful optimization geometry can be smaller than full state-space geometry. SU(2)-Haar is the clearest example: approximately three quarters of QFIM trace is absent from the ≤ 2 accessible metric, yet AQNG generalizes better on average across all four benchmarks. Conversely, the $U(1)$ control shows that reproducing the full-QNG direction is not sufficient for task success. This motivates the hierarchy

$$\boxed{\text{state geometry} \longrightarrow \text{measurement-accessible geometry} \longrightarrow \text{task-useful optimization geometry.}} \tag{16}$$

The first arrow is governed by measurement/readout contraction. The second additionally depends on loss, labels, data encoding, and finite optimization trajectory.

B. Limitations

The experiments use six-qubit noiseless statevector simulation, small binary datasets, one fixed split per dataset, and 20 optimization steps. They establish a controlled finite-size comparison, not an asymptotic scaling theorem or hardware advantage. AQNG is not guaranteed to outperform full QNG: RyRz-CZ on Digits is a clear counterexample. The timing comparison is also confounded by different metric simulator backends; a matched-backend study should separately count state preparations, circuit evaluations, shots, and classical linear-algebra cost. The present metric uses fixed diagonal readout probes through weight two; general POVMs, rotated bases, adaptive readouts, finite-shot covariance estimation, and noisy channels can alter accessibility. Finally, the $U(1)$ data do not identify the precise source of task failure. Encoding mismatch, conserved-sector expressivity, and readout/label alignment require targeted ablations.

VI. CONCLUSION

We introduced the accessible quantum natural gradient, which replaces the full quantum Fisher metric with the metric induced by measurement/readout-accessible tangent geometry. AQNG is motivated by the spectral identity $\mathbb{E}[I/F_Q] = \mathrm{Tr}(TC_Q)$ and is therefore an algorithmic application of accessible-tangent theory rather than merely a low-cost QFIM approximation.

Across four binary classification datasets, four six-qubit architectures, and 20 paired seeds per cell, AQNG is competitive with full QNG and improves mean accuracy in nine of twelve nonconserving settings. SU(2)-Haar is particularly informative: the accessible metric retains only about one quarter of QFIM trace while AQNG improves mean accuracy on all four tasks, with corrected significant loss gains on Iris and Wine. Digits/SU(2)-CNOT provides a third corrected significant loss gain. The symmetry-constrained $U(1)$ control remains at fixed chance-like accuracy under both optimizers despite strong direction alignment, separating natural-gradient fidelity from task utility.

These results suggest that variational quantum optimization need not reconstruct every locally distinguishable state-space direction. What matters is the geometry that survives the learning interface and is useful to the task. Establishing when this accessible geometry yields provable optimization or sample-complexity advantages is the central open problem raised by AQNG.

ACKNOWLEDGMENTS

The author thanks the open-source scientific Python and quantum-software communities. Computational results and aggregation files are available in the accompanying reproducibility repository [6].

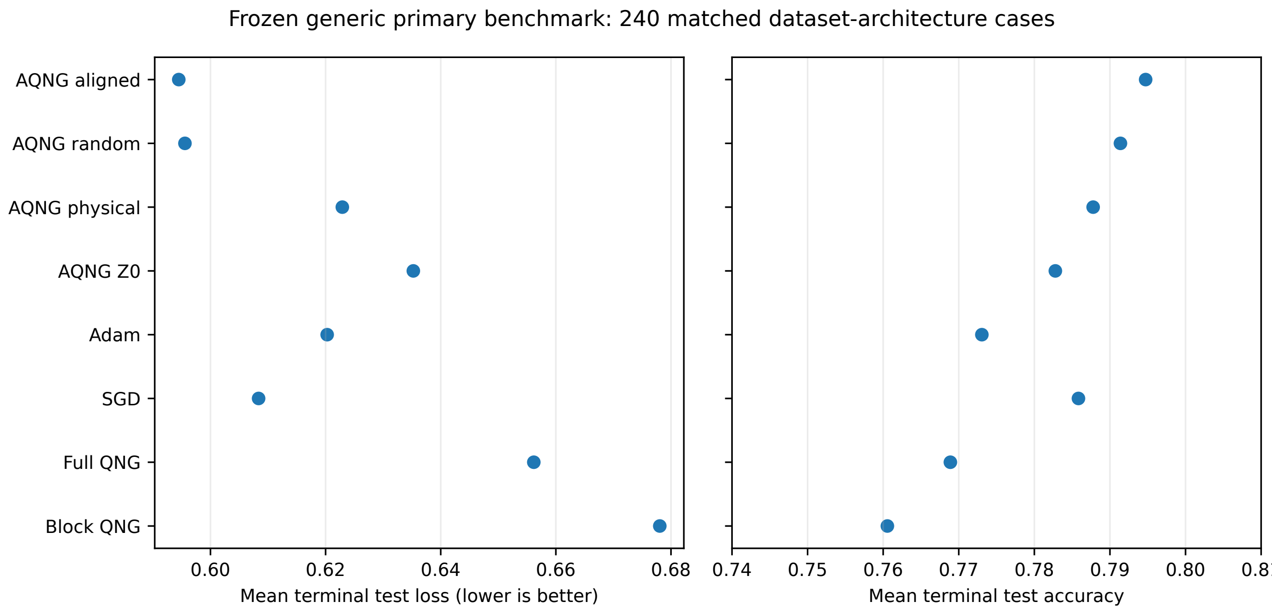

FIG. 2. Frozen generic primary benchmark pooled over the 240 nonconserving dataset–architecture cases. The primary protocol compares eight methods at matched data partitions and stochastic seeds. Aligned AQNG has the lowest pooled mean terminal test loss (0.5945) and highest pooled mean accuracy (0.7947), but random AQNG and SGD remain close. The pooled means are descriptive summaries; inferential claims use paired tests rather than treating all terminal rows as independent replicates.

DATA AND CODE AVAILABILITY

The complete experiment contains 80 dataset–seed jobs and 640 terminal result rows. Aggregate tables, per-seed results, metric diagnostics, and workflow metadata are stored under `results/paper/paper_main_v1/` in Ref. [6]. The released manifest records the exact workflow configuration and numerical validation tolerance used for reaggregation.

Appendix A: Exact AQNG construction used in the experiments

This appendix specifies the AQNG instance used for the reported benchmark so that the optimizer can be reconstructed from the experimental record. We distinguish the mathematical construction from run-time arguments: the former defines the accessible metric, whereas the latter controls its numerical estimation and refresh schedule.

1. Inputs and retained readout family

For a parameterized circuit $U(\boldsymbol{\theta})$ on n qubits and an encoded input x, AQNG takes (i) the current parameter vector $\boldsymbol{\theta} \in \mathbb{R}^p$, (ii) a metric minibatch B_G, (iii) a retained observable list $\mathcal{A} = \{O_a\}_{a=1}^r$, (iv) a damping coefficient λ, and (v) a numerical pseudoinverse cutoff `rcond`. The paper configuration uses the complete diagonal Pauli-Z family through weight two,

$$\mathcal{A}_{\leq 2} = \{Z_i\}_{i=1}^n \cup \{Z_i Z_j\}_{1 \leq i < j \leq n}. \tag{A1}$$

For $n = 6$ this gives $r = 6 + \binom{6}{2} = 21$ retained features. Two smaller spaces are used only as diagnostics: $A_1 = \operatorname{span}\{Z_0\}$ and the local space $\operatorname{span}\{Z_i\}_{i=1}^6$. They are not substituted for $\mathcal{A}_{\leq 2}$ in the main AQNG training runs.

For each $x \in B_G$, define the retained expectation vector

$$\boldsymbol{m}(x, \boldsymbol{\theta}) = \big(\langle O_1 \rangle, \ldots, \langle O_r \rangle\big)^{\mathsf{T}}, \tag{A2}$$

its parameter Jacobian $J_x = \partial \boldsymbol{m} / \partial \boldsymbol{\theta} \in \mathbb{R}^{r \times p}$, and the symmetrized feature covariance

$$(\Sigma_x)_{ab} = \frac{1}{2} \langle O_a O_b + O_b O_a \rangle - \langle O_a \rangle \langle O_b \rangle. \tag{A3}$$

Because the retained observables are diagonal Pauli-Z strings, they commute in the present implementation. The covariance-normalized score map is represented by $\Sigma_x^{+1/2} J_x$, with null covariance directions removed by the pseudoinverse. The per-example accessible Gram matrix is therefore

$$G_A(x) = J_x^{\mathsf{T}} \Sigma_x^{+} J_x, \tag{A4}$$

and the metric used by one refresh is the minibatch average

$$\widehat{G}_A(\boldsymbol{\theta}; B_G) = \frac{1}{|B_G|} \sum_{x \in B_G} G_A(x). \tag{A5}$$

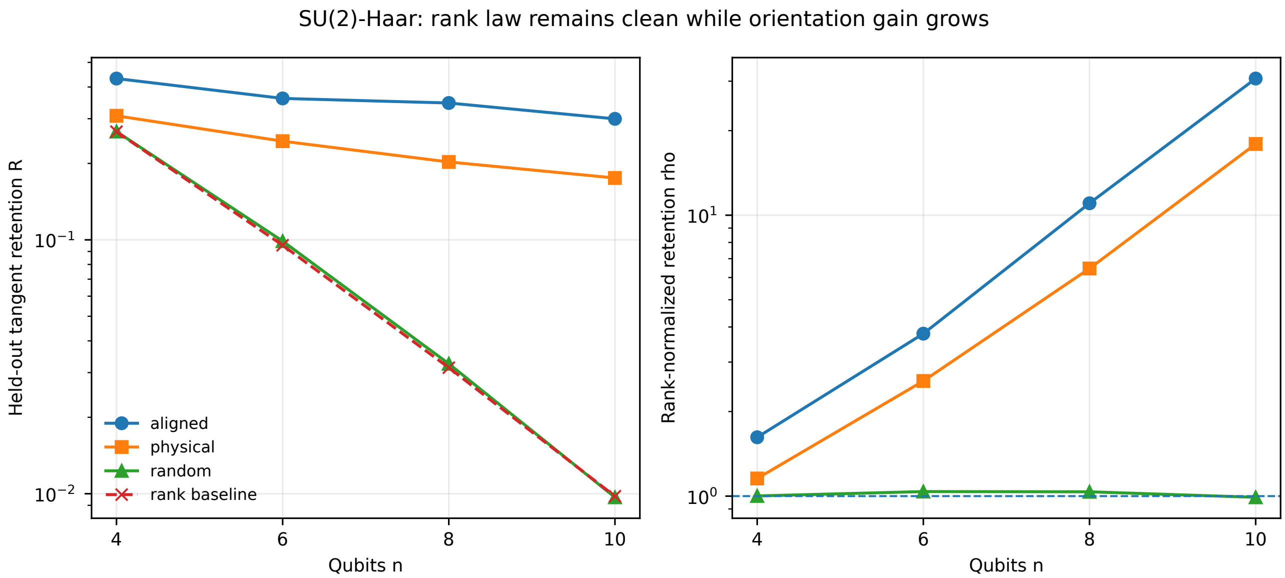

FIG. 3. Qubit scaling of SU(2)-Haar readout geometry at fixed readout order. Left: held-out tangent retention R for cross-fitted aligned, physical, and Haar-random equal-rank readouts, together with the rank-only baseline r/N. The baseline falls from 0.267 at $n = 4$ to 9.78×10^{-3} at $n = 10$, while the aligned readout still retains about 0.300 at $n = 10$. Right: rank-normalized retention ρ. The random control remains near $\rho = 1$, whereas the physical and aligned readouts rise to approximately 17.9 and 30.7 at $n = 10$. The growth is a task-conditioned orientation effect, not a claim that generic physical readouts violate the rank law.

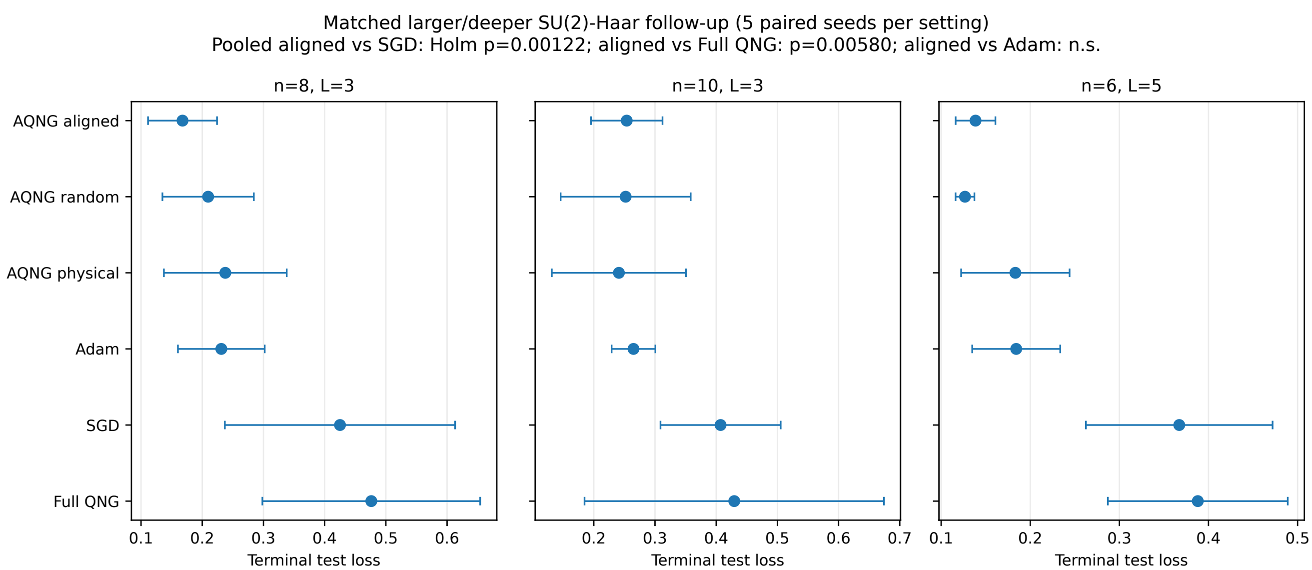

FIG. 4. Matched larger/deeper SU(2)-Haar follow-up. Error bars show the standard deviation across five paired seeds within each setting. Pooled over the 15 matched cells, aligned AQNG beats SGD in 15/15 cases with Holm-adjusted $p = 1.22 \times 10^{-3}$ and full QNG in 14/15 with $p = 5.80 \times 10^{-3}$. The aligned–Adam difference is smaller (10/15 wins) and is not statistically resolved. The follow-up therefore supports a robust advantage over SGD and the tested full-QNG baseline in these regimes, but not universal superiority over Adam.

Equations (A3)–(A5) are the finite-feature realization of the pullback in Eq. (12): the Jacobian maps parameter motion into retained readout motion, while Σ_x^+ supplies the covariance/Fisher normalization on that retained motion.

2. Update rule and metric refresh

Let B_L be the loss minibatch and

$$\boldsymbol{g}_t = \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t; B_L) \tag{A6}$$

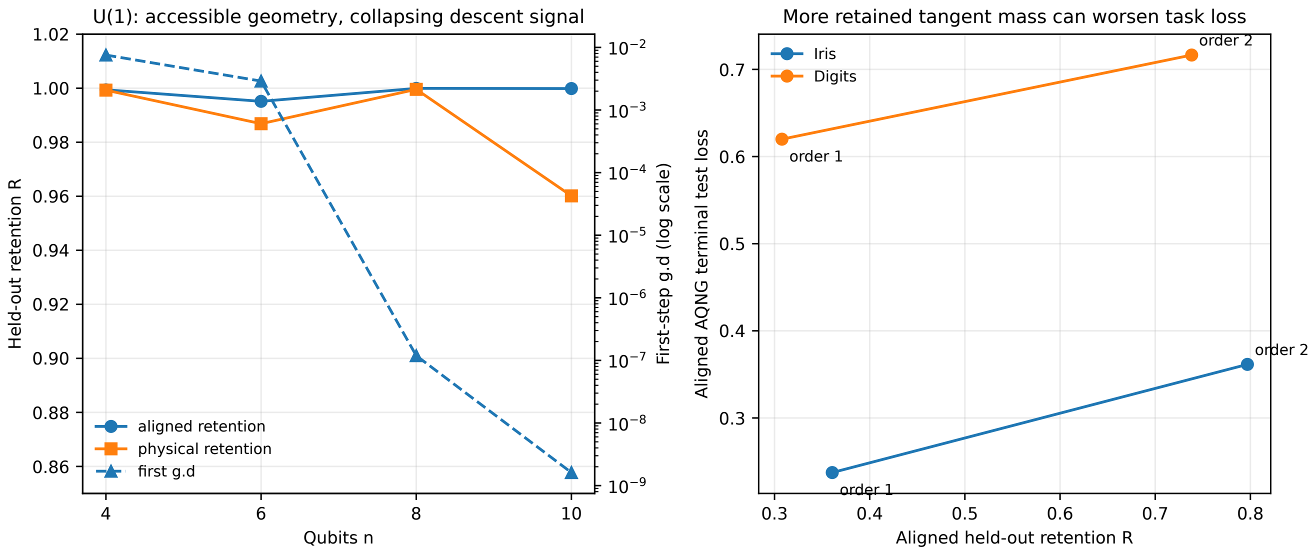

FIG. 5. Accessibility is not trainability. Left: in the half-filled $U(1)$ family, physical and cross-fitted aligned readouts retain nearly all held-out tangent mass over the tested scaling range, while the first-step $g^{\mathsf{T}}d$ diagnostic collapses by many orders of magnitude and terminal accuracy remains chance-like. Right: for SU(2)-Haar, increasing the readout from order one to order two moves both Iris and Digits toward substantially larger aligned retention but also larger terminal aligned-AQNG loss. Retention is therefore a geometric accessibility diagnostic, not a monotone task-performance objective.

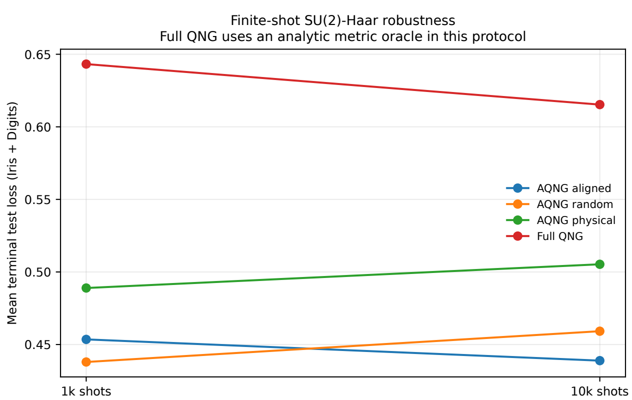

FIG. 6. Finite-shot SU(2)-Haar comparison averaged over the Iris and Digits finite-shot cells. AQNG remains competitive at both 10^3 and 10^4 shots in the tested protocol, without a monotone terminal-loss improvement as the shot count increases. Full QNG is included as an optimization reference but uses an analytic metric oracle in these finite-shot runs; the figure must therefore not be interpreted as a hardware-fair resource comparison.

its ordinary parameter gradient. At a metric-refresh step, AQNG constructs $\widehat{G}_{A,t}$ from Eq. (A5); otherwise it reuses the most recently constructed matrix. The implemented preconditioned direction is obtained from the damped linear system

$$(\widehat{G}_{A,t} + \lambda I)\boldsymbol{d}_t = \boldsymbol{g}_t, \qquad \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \boldsymbol{d}_t. \tag{A7}$$

A pseudoinverse/SVD cutoff `rcond` $= 10^{-8}$ is used for numerically singular covariance/metric directions. Damping is applied before the update solve; it regularizes small accessible eigenvalues and prevents null or nearly null retained directions from producing arbitrarily large parameter steps.

The metric is deliberately not rebuilt on every optimization step. With `metric_every`$=$ 2, the 20-step benchmark performs 10 metric builds. With `metric_batch`$=$ 4, this corresponds to 40 metric examples per method and seed. The loss gradient uses `loss_batch`$=$ 8 at each of 20 steps, for 160 loss examples per method and seed. These logical workloads are matched between AQNG and full QNG; the wall-clock comparison in the main text remains backend-specific.

3. Arguments used in the paper run

Table II records the optimizer and experiment arguments preserved by the run manifest and per-seed configuration files. The stochastic seed does *not* change the train/test split. It controls parameter initialization, minibatch scheduling, and, for `su2_haar`, the fixed Haar entangling instance. Thus the 20 repetitions probe training/circuit stochasticity on one fixed split per dataset.

4. Reference algorithm

The complete AQNG iteration can be summarized as follows:

1. Fix the retained observable family $\mathcal{A}_{\leq 2}$ and initialize $\boldsymbol{\theta}_0$.

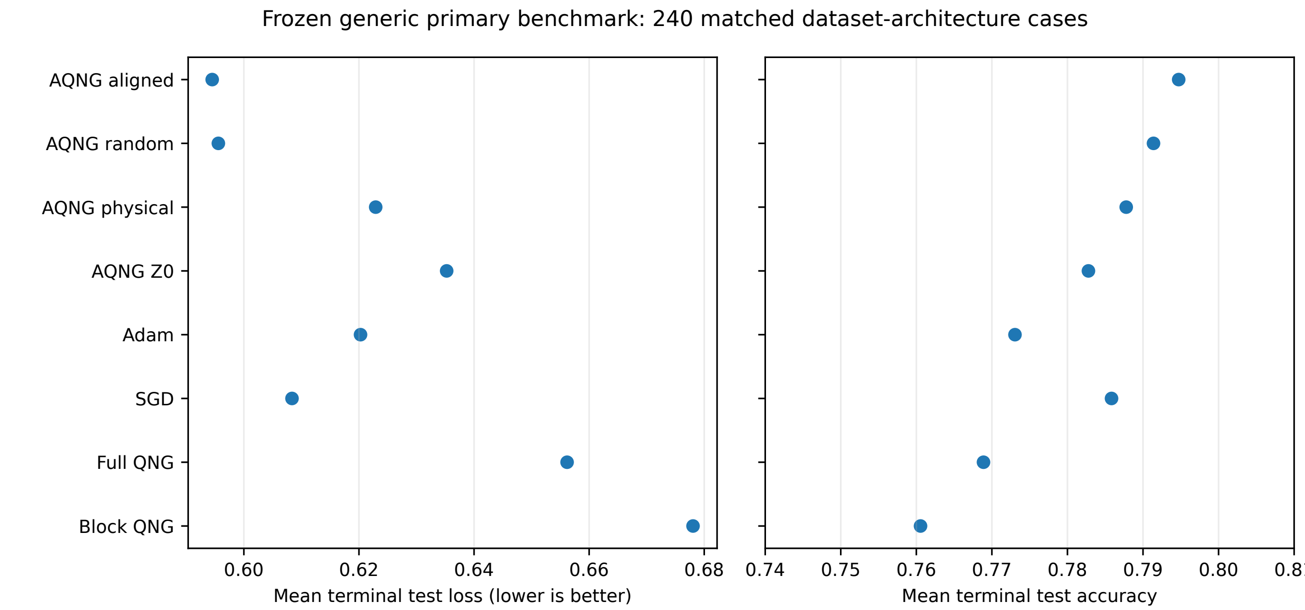

FIG. 7. Frozen generic primary benchmark pooled over the 240 nonconserving dataset–architecture cases. The primary protocol compares eight methods at matched data partitions and stochastic seeds. Aligned AQNG has the lowest pooled mean terminal test loss (0.5945) and highest pooled mean accuracy (0.7947), but random AQNG and SGD remain close. The pooled means are descriptive summaries; inferential claims use paired tests rather than treating all terminal rows as independent replicates.

TABLE II. Arguments of the AQNG benchmark instance. "Role" describes how each argument enters the construction or experimental protocol.

Argument	Paper value	Role
`n_qubits`	6	Circuit width; gives 21 $Z_{\leq 2}$ readout features
`n_layers`	3	Variational circuit depth parameter
`steps`	20	Number of parameter updates
`learning_rate` / `lr`	0.03	η in Eq. (A7)
`damping` / `lam`	10^{-3}	Tikhonov damping λ
`rcond`	10^{-8}	Pseudoinverse/SVD relative cutoff
`loss_batch`	8	Examples used for each loss gradient
`metric_batch`	4	Examples averaged in each metric refresh
`metric_every`	2	Rebuild metric every second optimization step
`readout`	$Z_{\leq 2}$	All Z_i and $Z_i Z_j$ retained observables
`n_samples`	80	Dataset subset size used in each benchmark
`subset_seed`	1729	Fixed seed selecting the dataset subset
`split_seed`	314159	Fixed train/test split seed
`seed`	0–19	Initialization, minibatch schedule, and fixed Haar instance

2. At step t, draw the paired loss minibatch B_L and evaluate $\boldsymbol{g}_t = \nabla L(\boldsymbol{\theta}_t; B_L)$.

3. If t is a refresh step, draw B_G; for every $x \in B_G$, evaluate $\boldsymbol{m}(x, \boldsymbol{\theta}_t)$, J_x, and Σ_x, then form $G_A(x) = J_x^{\mathsf{T}} \Sigma_x^+ J_x$ and average the four matrices. Otherwise reuse the preceding $\widehat{G}_A$.

4. Solve $(\widehat{G}_A + \lambda I)\boldsymbol{d}_t = \boldsymbol{g}_t$ using the stated numerical cutoff.

5. Update $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \boldsymbol{d}_t$.

Full QNG follows the same loss minibatches, metric minibatch size, refresh cadence, damping, and learning rate, but replaces $\widehat{G}_A$ by the full pure-state QFIM. This pairing is why the initial gradient is identical up to floating-point precision and why differences can be attributed to the preconditioning geometry rather than to a different stochastic workload.

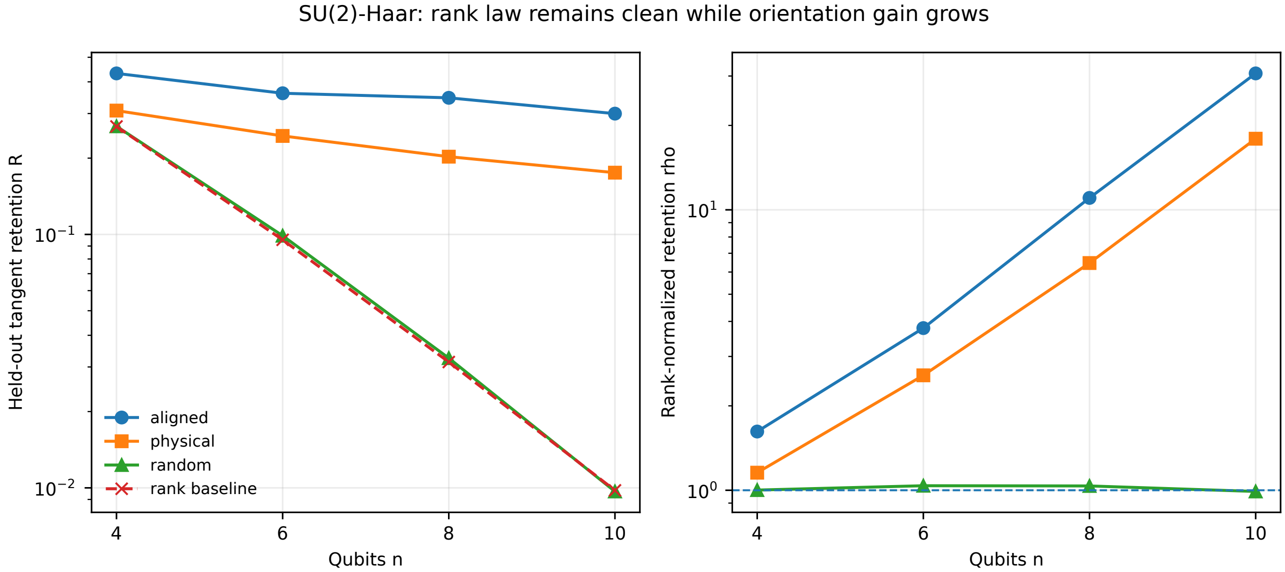

FIG. 8. Qubit scaling of SU(2)-Haar readout geometry at fixed readout order. Left: held-out tangent retention R for cross-fitted aligned, physical, and Haar-random equal-rank readouts, together with the rank-only baseline r/N. The baseline falls from 0.267 at $n = 4$ to 9.78×10^{-3} at $n = 10$, while the aligned readout still retains about 0.300 at $n = 10$. Right: rank-normalized retention ρ. The random control remains near $\rho = 1$, whereas the physical and aligned readouts rise to approximately 17.9 and 30.7 at $n = 10$. The growth is a task-conditioned orientation effect, not a claim that generic physical readouts violate the rank law.

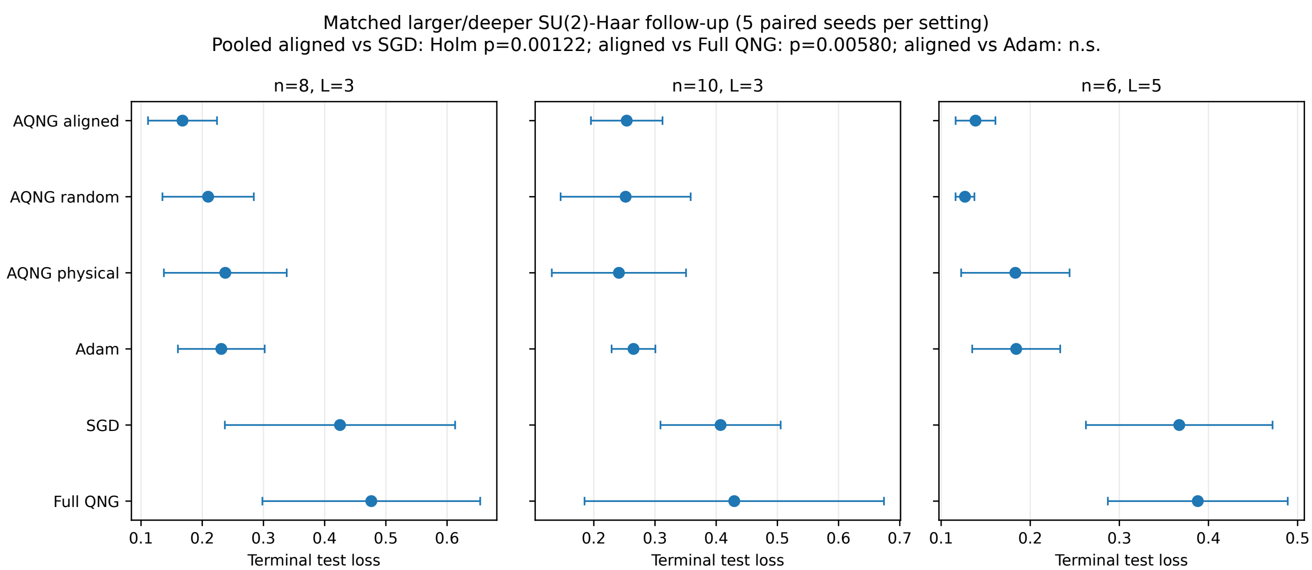

FIG. 9. Matched larger/deeper SU(2)-Haar follow-up. Error bars show the standard deviation across five paired seeds within each setting. Pooled over the 15 matched cells, aligned AQNG beats SGD in 15/15 cases with Holm-adjusted $p = 1.22 \times 10^{-3}$ and full QNG in 14/15 with $p = 5.80 \times 10^{-3}$. The aligned–Adam difference is smaller (10/15 wins) and is not statistically resolved. The follow-up therefore supports a robust advantage over SGD and the tested full-QNG baseline in these regimes, but not universal superiority over Adam.

5. Diagnostics and consistency checks

At the initial paired point the implementation additionally constructs G_{A_1}, G_{local}, $G_{\leq 2}$, and G_Q. It records their ranks and traces, the ratios $\mathrm{Tr}(G_{\mathcal{A}})/\,\mathrm{Tr}(G_Q)$, and the cosine between each preconditioned direction and the full-QNG direction. It also checks the nested Loewner or-

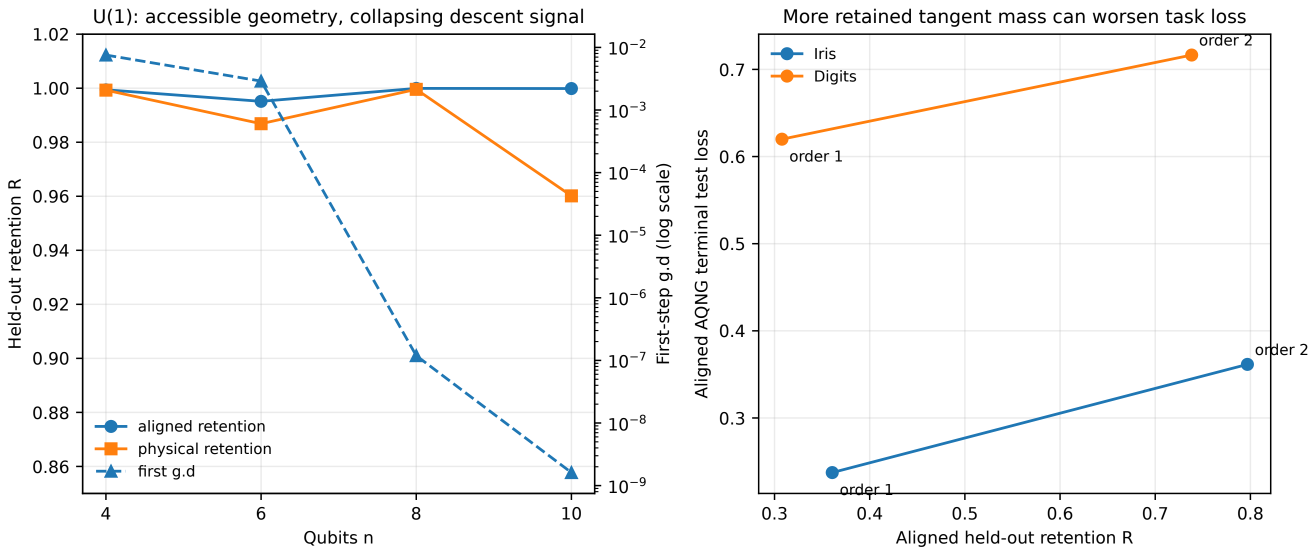

FIG. 10. Accessibility is not trainability. Left: in the half-filled $U(1)$ family, physical and cross-fitted aligned readouts retain nearly all held-out tangent mass over the tested scaling range, while the first-step $g^{\mathsf{T}}d$ diagnostic collapses by many orders of magnitude and terminal accuracy remains chance-like. Right: for SU(2)-Haar, increasing the readout from order one to order two moves both Iris and Digits toward substantially larger aligned retention but also larger terminal aligned-AQNG loss. Retention is therefore a geometric accessibility diagnostic, not a monotone task-performance objective.

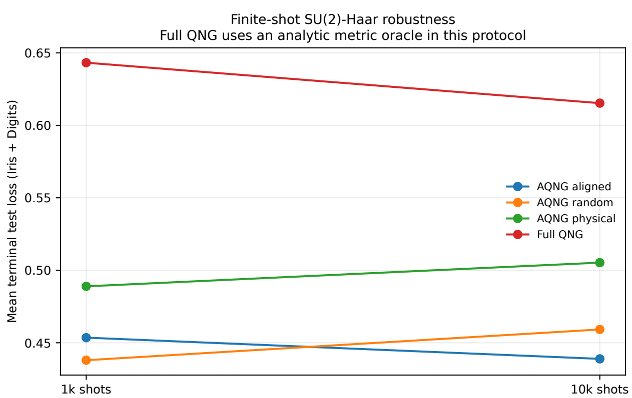

FIG. 11. Finite-shot SU(2)-Haar comparison averaged over the Iris and Digits finite-shot cells. AQNG remains competitive at both 10^3 and 10^4 shots in the tested protocol, without a monotone terminal-loss improvement as the shot count increases. Full QNG is included as an optimization reference but uses an analytic metric oracle in these finite-shot runs; the figure must therefore not be interpreted as a hardware-fair resource comparison.

dering

$$G_{A_1} \preceq G_{\text{local}} \preceq G_{\leq 2} \preceq G_Q. \tag{A8}$$

The production run used a scale-aware numerical validation tolerance of order 10^{-5} in the pre-run smoke gate. The largest observed nesting residual was 8.706×10^{-6} in Wine/$U(1)$; no training datum or terminal statistic was modified during reaggregation.

6. Scope of the implementation description

The equations above specify the accessible metric and all numerical arguments recorded for the reported run. They should not be read as requiring diagonal Pauli readout in general. For a different measurement/readout interface, $\boldsymbol{m}$, J, and Σ are replaced by the corresponding retained score coordinates; the abstract construction remains the pullback of the accessible tangent map. Likewise, `metric_batch`, `metric_every`, λ, and `rcond` are numerical hyperparameters rather than defining properties of AQNG itself.

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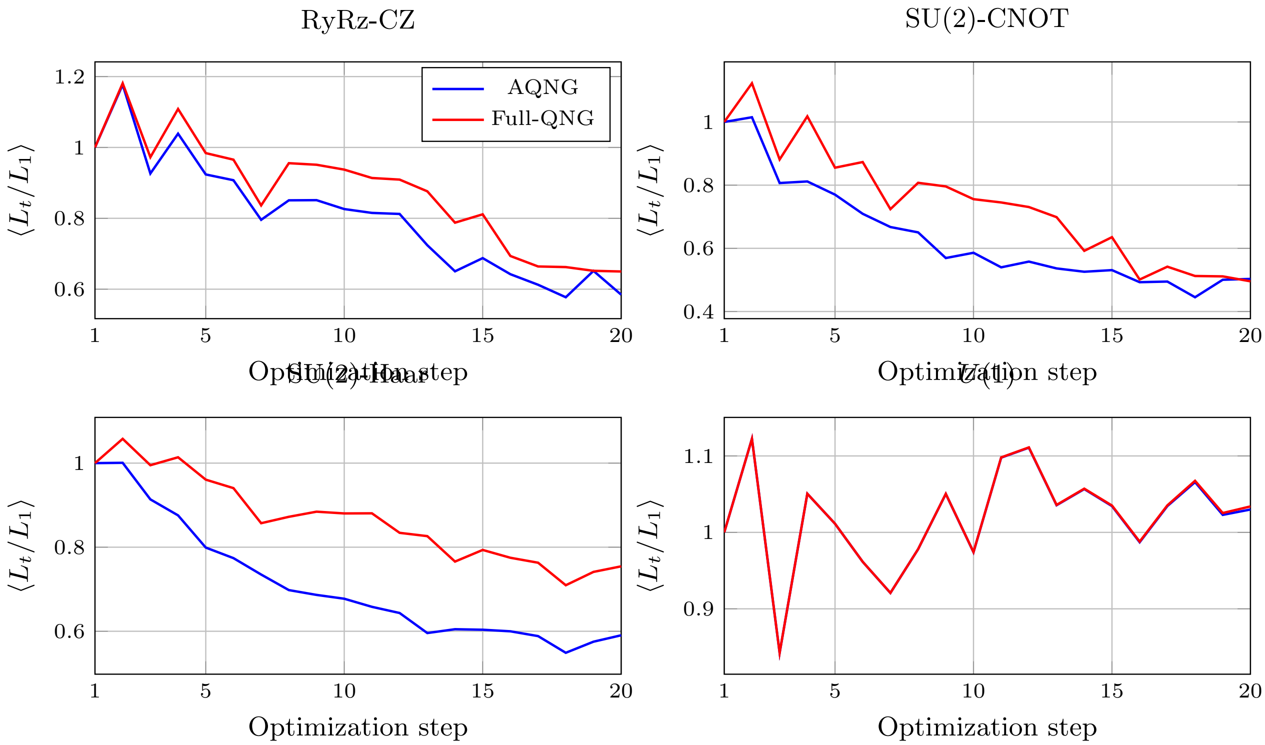

FIG. 12. Training dynamics from the raw per-step curve logs. For each architecture and optimizer, the minibatch loss is normalized by its first-step value within each run and then averaged over all four datasets and 20 seeds per dataset (80 trajectories per curve). The curves therefore show optimizer dynamics rather than terminal test performance. Per-step classification accuracy was not logged in the original run; terminal test accuracy is reported separately in Fig. **??**.

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